Analysis; Implementation Support; Evaluation Technique; Help and Documentation: Requirements of user support, Approaches to user support, Intelligent help Systems.

Groupware: Group wave systems, Meeting and Decision support systems, Framework for Grouware.

CSCW Issues and Theory: Face to Face Communication, conversation.

Multi-sensory Systems : Usable sensory Inputs, speech in the interface, Handwriting Recognition; Text Hypertext and Hypermedia; Gesture Recognition, Computer Vision, Application of Multimedia Systems.

Recommended Texts:

1. A. Dix et al., *Human-Computer Interaction*, 3rd ed. Pearson, 2003

SWE 4839 Big Data Analysis Credit: 3.00

Transition from relational database to big data & from data-mining to big data, Business aspect of big data, Characteristics of big data, Distributed File Systems, Map Reduce, Finding Similar items, Link Analysis algorithms, Similarity measure algorithms, Data Filtering algorithms, Introduction to NoSQL databases, Introduction to Hadoop platform.

Recommended Texts:

1. J. Leskovec et al., *Mining of Massive Datasets*, 3rd ed. Cambridge University Press, 2019
2. T. White, *Hadoop: The Definitive Guide, Storage and Analysis at Internet Scale*, 4th ed. O'Reilly Media, 2015

SWE 4841 Natural Language Processing Credit: 3.00

Classical Approaches to Natural Language Processing, Text Preprocessing, Lexical Analysis, Syntactic Parsing, Semantic Analysis, Natural Language Generation, Corpus Creation, Part-of-Speech Tagging, Information Extraction, Statistical Parsing, Multiword Expression, Normalized Web Distance and Word Similarity, Word Sense Disambiguation, Machine Translation, Applications of NLP, Deep Learning for NLP.

Recommended Texts:

1. D. Jurafsky and J. Martin, *Speech & language processing*, 2nd ed. Prentice Hall, 2008
2. C. Manning and H. Schutze, *Foundations of Statistical Natural Language Processing*, 1st ed. The MIT press, 1999

SWE 4843 Concurrent and Parallel Programming Credit: 3.00

This course introduces the design, development and debugging of parallel programs. It will build on the concurrency concepts gained from the Operating Systems module. It covers concepts and modeling tools for specifying and reasoning (about the properties of) concurrent systems and parallel programs. It also covers principles of performance analysis, synchronous and asynchronous parallel programming, and engineering concurrent systems and parallel programs. The topics includes: Concurrency Basics; From Concurrency to Concurrent Programming; Basic exposure to Multi-threaded Java; Threads and concurrent execution; Managing concurrency via locks and shared objects; Monitors as a concurrency control mechanism; Deadlocks in concurrent systems; Parallel programming using MPI (point-to-point communication, collective communication, management of communicators).

Recommended Texts:

1. J. Magee and J. Kramer, *Concurrency: State Models and Java Programs*, 2nd ed. Wiley, 2006
2. C. Lin and L. Snyder, *Principles of Parallel Programming*, 1st ed. Pearson, 2008

SWE 4845 E-Commerce Credit: 3.00

E-commerce Business Models and Concepts: Identify the key components of e-commerce business models, B2C business models, and major B2B business models, Recognize business models in other emerging areas of e-commerce, key business concepts and strategies applicable to e-commerce.

The Internet and World Wide Web: E-commerce Infrastructure: The origins of the Internet, Key technology concepts behind the Internet, Role of Internet protocols and utility programs, Current structure of the Internet, How the World Wide Web works, How Internet and Web features and services support e-commerce.

E-commerce Marketing concept: Identify the key features of the Internet audience, Basic concepts of consumer behavior and purchasing decisions, Understanding how consumers behave online, Basic marketing concepts needed to understand Internet marketing, Main technologies that support online marketing.

Ethical, Social, and Political Issues in E-commerce: Main ethical, social, and political issues raised by e-commerce, A process for analyzing ethical dilemmas, Basic concepts related to privacy, Practices of e-commerce companies that threaten privacy, Different methods used to protect online privacy, Major public safety and welfare issues raised by e-commerce.

Online Security and Payment Systems: Scope of e-commerce crime and security problems, Key dimensions of e-commerce security, Key security threats in the e-commerce environment, How technology helps protect the security of messages sent over the Internet, Tools used to establish secure Internet communications channels,